



louis.hebrard@ens-paris-saclay.fr



louheb.github.io/cv

Skills

Computer:

(years of experience)

Python —●●●●+

Matlab —●●●●●

ImageJ/FIJI —●●●●

Abaqus —●●●●●

Gmsh —●●●●●

Cast3M —●

SolidEdge —●●●●

Catia V5 —●●

Solidworks —●

Latex —●●●●+

LAMMPS —●

Excel —●●●●+

Photoshop —●●●●+

Experiments:

Lab. mechanical tests

SEM

EBS

Polishing & Sample prep.

X-ray μ -tomography

Synchrotron experiments

Neutron diffraction

Digital Image Correlation

Languages:

French - Native Speaker

English - C1 Advanced

German - B1 Level

Spanish - Basic Knowledge

French driving licence

(since 2015)

Interests

- Basketball
- Arts (drawing, painting)
- Social problematics

Louis HÉBRARD

- Material Sciences & Mechanical Engineering -

Research projects:

Since 2025: Postdoctoral fellowship

Component Safety Department (Integrity of Welded Joints) - Fatigue group
Bundesanstalt für Materialforschung und -prüfung - Berlin

- High temperature fatigue experiments on VDM® Alloy 780.
- Mechanical tests using different acquisition methods (DIC, IR acquisition, etc.).
- Neutron diffraction analyses

2020 - 2023: PhD thesis

Study on the effect of the environment on internal fatigue crack propagation in Ti64
Matéis Lab - INSA Lyon

JY. Buffiere - T. Palin-Luc (I2M Bordeaux) - N. Ranc (PIMM Paris)

- Synchrotron experiment on Ti-6Al-4V very high cycle fatigue crack propagation
- Effect of environment: comparison of crack propagation in air and in vacuum

2020: Research internship (6 months)

In-situ synchrotron tomography of laser powder bed fusion additive manufacturing
Simap - Grenoble & PIMM - Arts et Métiers - Paris

P. Lhuissier - G. Martin - P. Peyre - F. Coste

- Synchrotron tomography adapted exp. device to study LPBF impacting parameters

2018 - 2019: Engineering internship (6 months)

Creation of a fatigue test bench for a humanoid-robot foot
Wandercraft - Paris

2017 - 2018: Research internship (10 months)

Numerical simulations at an atomic scale of the solute segregation toward grain boundaries in an Al-Mg alloy

The University of British Columbia - Vancouver

C.W. Sinclair - Microstructure group

- Modelling of grain boundaries and segregation with hybrid Molecular Dynamic Method
- Post-processing of the numerical data to build free energy diagrams

2017: Engineering internship (14 weeks)

Evaluation of the control loads on a H175 helicopter
Airbus Helicopters - Marignane

Publications:

2025 - Hébrard et al. - International Journal of Fatigue - Deformation mechanisms at the tip of internal fatigue cracks in vacuum and in the presence of an air environment in a Ti alloy - doi.org/10.1016/j.actamat.2023.119608

2024 - Ball et al. - Acta Materialia - doi.org/10.1016/j.actamat.2023.119608

2023 - Hebrard et al. - Materials Science and Engineering: A - Environment effect on internal fatigue crack propagation studied with in-situ X-ray microtomography doi.org/10.1016/j.msea.2023.145462

2022 - Lhuissier et al. - Review of Scientific Instruments doi.org/10.1063/5.0090623

2018 - Daghia et al. - Composites Part A: Applied Science and Manufacturing doi.org/10.1016/j.compositesa.2018.07.020

Teaching experience:

2020 - 2022: Mechanical system conception, Technical drawing and Computed-aided design (undergraduates from INSA Lyon)